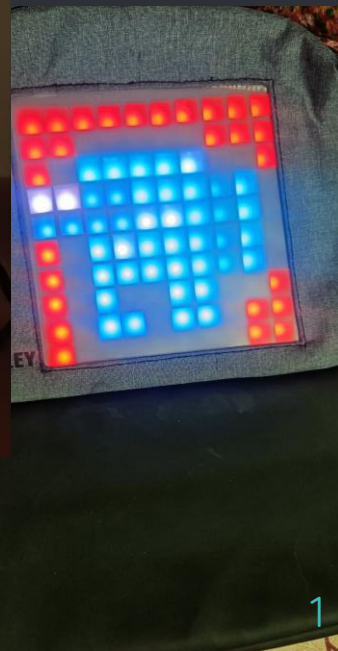
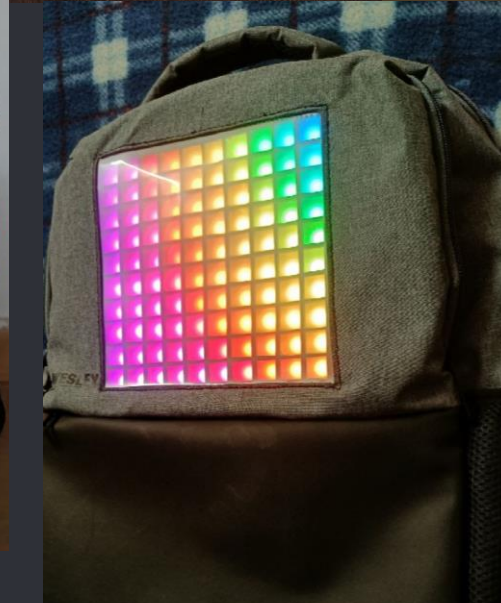
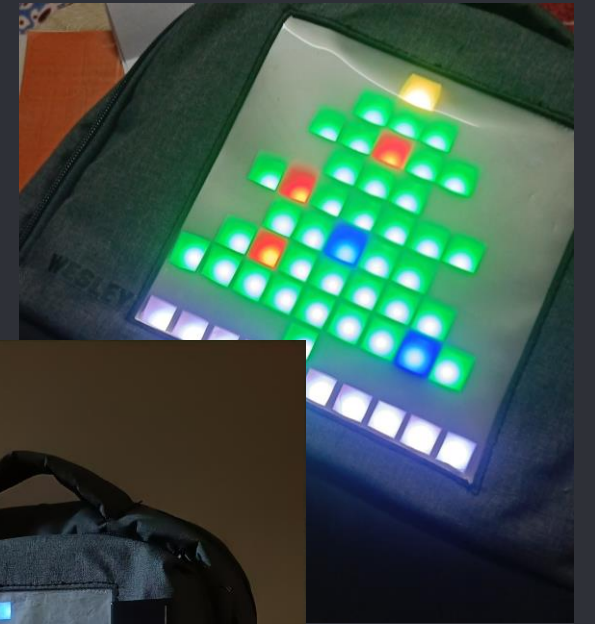
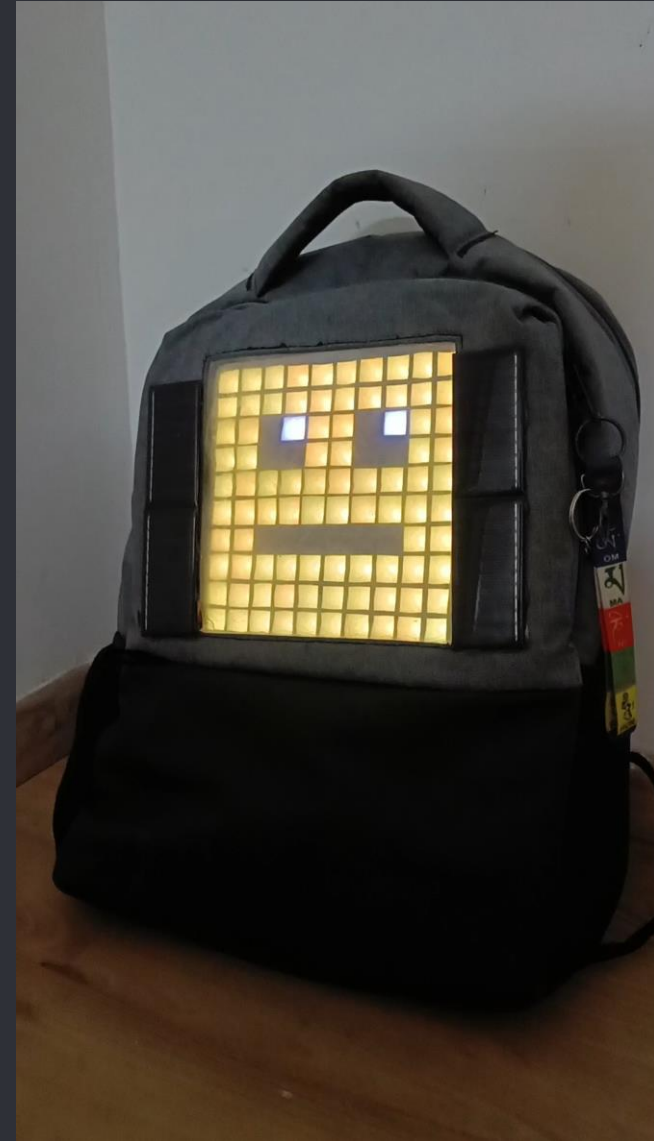


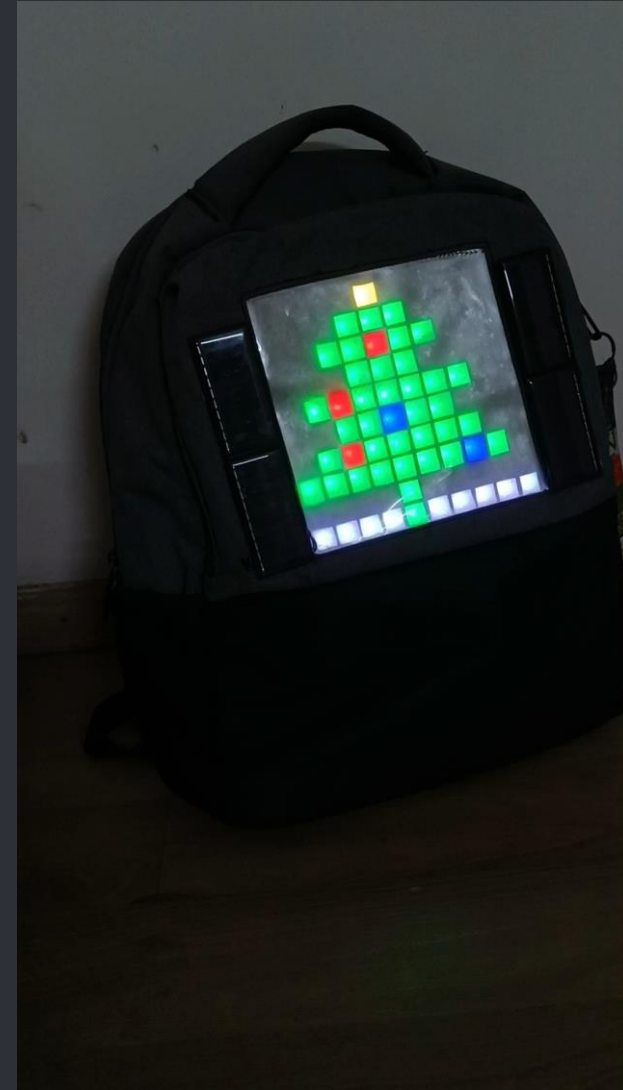
PROTOTYPE



● PROTOTYPE

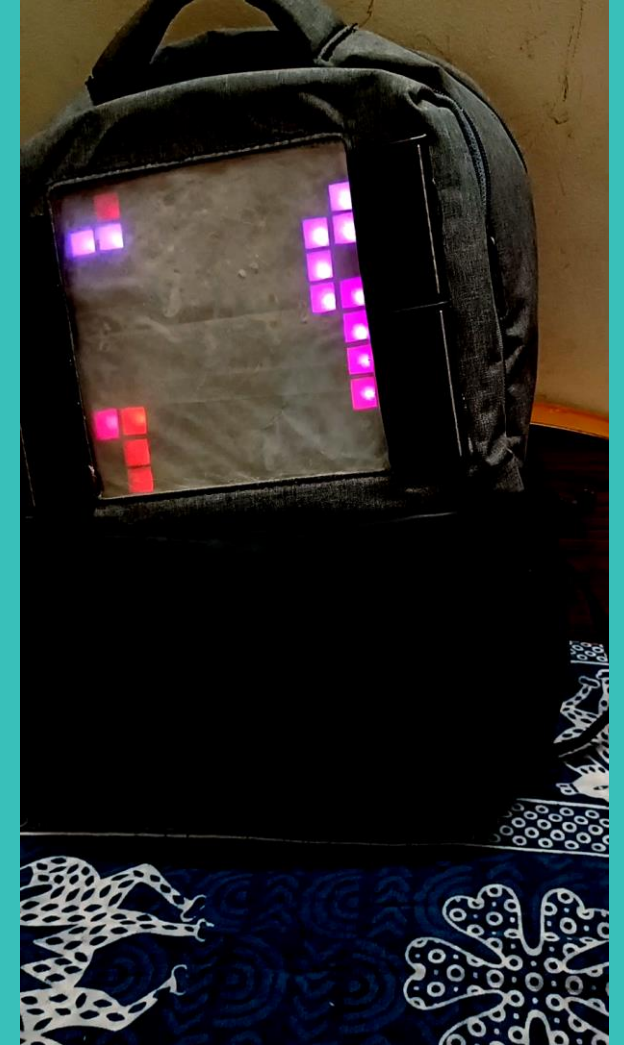


● PROTOTYPE

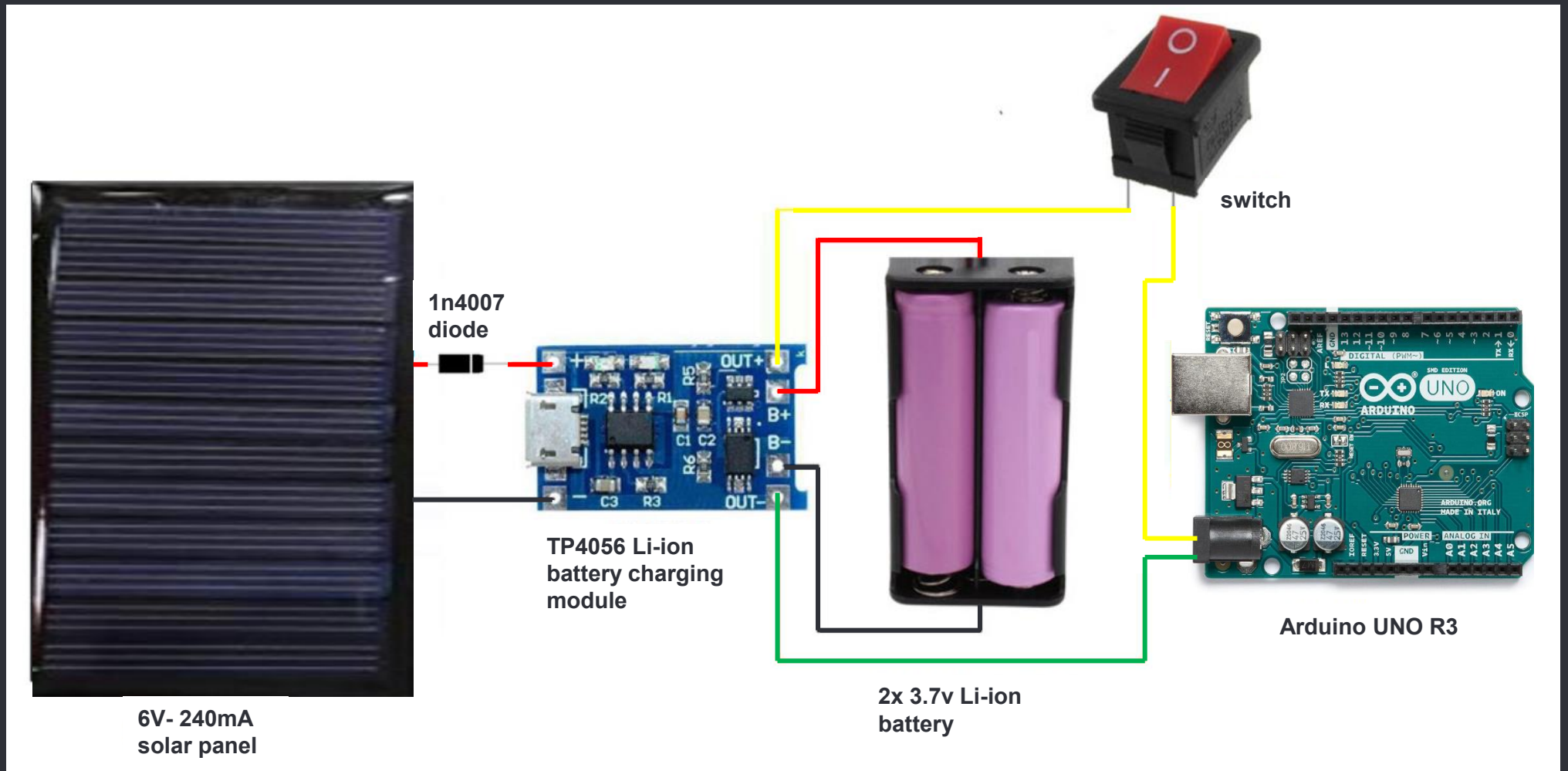


APPLICATIONS

- Entertainment on go.
- Can be used for safety indications in case of low visibility.
- Built in charging port.
- Can used for displaying messages.

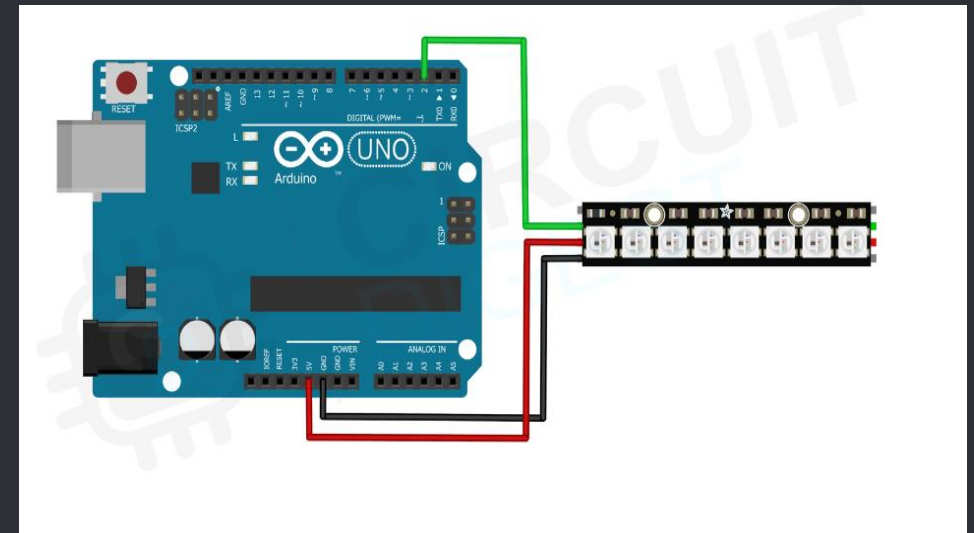
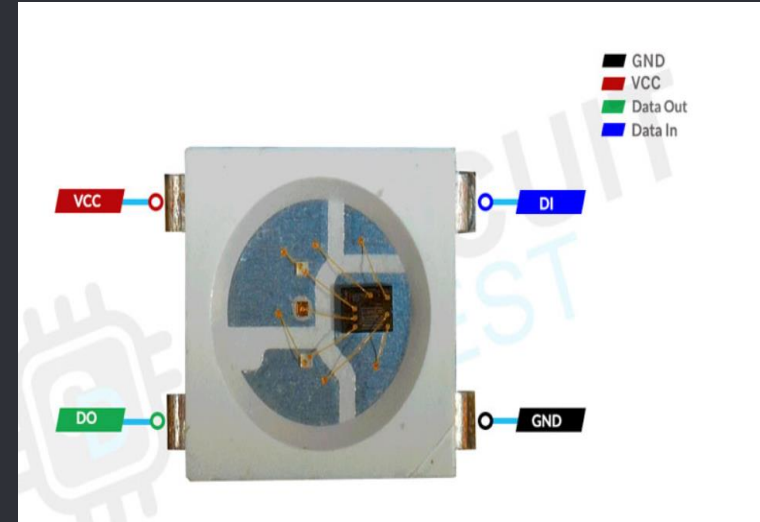


● CIRCUIT DIAGRAM



● RGB LED USED

- No. of LEDs in 1 Strip - 10
 - No. of Strips used – 10
 - (10x10 Grid)
-
- Maximum current consumption :
 - RGB (White light) :
 - $20\text{mA} + 20\text{mA} + 20\text{mA} = 60\text{mA}$



ABSTRACT

The solar-powered RGB LED matrix backpack is an innovative and sustainable solution to traditional lighting systems. With a 10x10 array of RGB LEDs, the system can be programmed to display various patterns, animations, and messages, making it an ideal accessory for outdoor events, festivals, and camping trips. The system is powered by a solar panel that charges the battery during the day, making it an environmentally friendly and sustainable option.

The backpack is designed to be lightweight, weather-resistant, and able to withstand outdoor conditions. The LED matrix is incorporated into the front panel of the bag, providing a unique and customizable display. The microcontroller controls the LED matrix and communicates with the solar charge controller to manage the charging and discharging of the battery. Multiple units can be connected to create larger displays, making it a scalable option for a range of applications.

This project aims to showcase the potential of solar-powered LED lighting systems as a sustainable and creative alternative to traditional lighting systems. Overall, this project demonstrates the importance of sustainability and innovation in creating a better future for our planet.

LITERATURE SURVEY

- https://www.researchgate.net/publication/338195210_Design_of_Solar_Powered_Charging_Backpack
- https://www.researchgate.net/publication/324676600_Design_of_Solar_Powered_Charging_Backpack
- <https://circuitdigest.com/microcontroller-projects/interfacing-neopixel-led-strip-with-arduino>
- <https://learn.adafruit.com/rgb-led-strips/current-draw>



● WHY TO USE SOLAR POWER?

1. Reliable, Cost effective and environment friendly resources
2. Saves money in the long run, Low Maintenance
3. Diverse Applications and usage
4. Flexibility, Renewable sources

THANKS!

